

CLAPC: A Hybrid CNN-LSTM Architecture for Automated Passenger Counting From Video Streams in Public Transport

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This work was supported by the Federal Ministry for Economic Affairs and Energy under Grant KK5060705MS2.

ABSTRACT Information on passenger occupancy in public transport is crucial for service providers to enable accurate revenue sharing within transportation networks. This requires reliable methods to count boarding and alighting passengers and to differentiate between passenger categories, such as those requiring a valid ticket. In previous work, a recurrent Long Short-Term Memory (LSTM) network (NAPC) was applied to depth videos captured by time-of-flight sensors mounted above train doors to perform passenger counting and height-based differentiation. In this work, we investigate whether this approach can be transferred to 2D gray-scale video data recorded from the same overhead perspective, where explicit depth information is unavailable. To this end, we extend the LSTM architecture by integrating a convolutional neural network, resulting in the CLAPC model, which improves the extraction of image features. Additionally, region-of-interest cropping and spatial down-sampling are explored to better address privacy concerns, and the training procedure is refined to reduce undercounting. Applying CLAPC to depth data improves counting accuracy compared to NAPC. Moreover, high accuracies are achieved on 2D video data despite the absence of explicit depth information. The highest performance is obtained using down-sampled door-path crops, yielding mean accuracies of 99.73% for boarding and 99.87% for alighting passengers. On depth data, accuracies of 99.79% and 99.97% are achieved for boarding and alighting, respectively. These results meet the $\pm 1\%$ accuracy requirement of VDV Guideline 457. The 95% confidence intervals from an equivalence test lie within the $\pm 1\%$ bounds, which confirms that CLAPC is statistically robust.

INDEX TERMS Intelligent transportation, automatic passenger counting (APC), long short-term memory (LSTM), convolutional neural network (CNN), boarding and alighting passenger counting, RGB video data.

I. INTRODUCTION

IN MODERN public transportation systems, accurate and reliable automatic passenger counting (APC) is essential for optimizing operations, improving service quality, and enabling data-driven decision-making. Passenger flow information supports demand-responsive scheduling, crowd management, and capacity planning [1], and also contributes to long-term infrastructure investment strategies. In regional networks where multiple operators jointly provide services, such as metropolitan areas and their surroundings, APC

further plays a crucial role in ensuring fair and transparent revenue sharing between operators. For this purpose, detailed and precise data on passenger boarding and alighting events is indispensable. When integrated with intelligent transport system or mobile applications, APC data can provide real-time and historical occupancy information, which can be used to enable passengers to assess seat availability and crowding levels before boarding and to choose less crowded routes, vehicles, or travel times [2], [3].

The history of automated passenger counting dates back to the mid-1970s [4] and is closely related to research in people detection ([5], [6], [7]), crowd density estimation ([8], [9]), and pedestrian flow analysis ([10], [11]). Advances in hardware technology, combined with rapid progress in

The review of this article was arranged by Associate Editor Vesna S ešumČavič.

machine learning, have since enabled the application of neural networks to a wide range of passenger counting tasks. Recent reviews highlight the diversity of APC technologies and emphasize the increasing role of computer vision and deep learning in operational fleets [12].

In video-based APC systems, detection-tracking-counting (DTC) pipelines are widely employed. State-of-the-art object detection methods include convolutional neural networks (CNN), variants of You Only Look Once (YOLO [13]), the Single Shot Multibox Detector (SSD [14]), and region-based approaches such as Regions with CNN Features (R-CNN [15]), Fast R-CNN [16], and Faster R-CNN [17]. These detectors are typically pretrained on large image datasets and then applied to video frames, where they are combined with multi-object tracking (MOT) algorithms such as Simple Online and Realtime Tracking (SORT [18]) or its enhanced variant incorporating a deep association metric (DeepSORT [19]).

For the counting stage, detection-tracking-counting pipelines typically rely on heuristics such as line-of-interest (LOI) and region-of-interest (ROI) strategies. For example, Kim et al. [20] introduced two reference lines and classified passenger behavior based on movement direction: a trajectory crossing from line 1 to line 2 was interpreted as boarding, whereas the reverse crossing was considered alighting. Rawat et al. [21], [22] adopted a ROI-based method. In their framework, where the door was not directly visible, ROIs were defined in proximity to the door area, and boarding or alighting events were inferred from the directional movement of passengers between these ROIs.

Most video-based APCs have been developed for bus-mounted cameras or evaluated on video data captured by bus-mounted cameras. Liu et al. [23] applied a CNN for head detection and incorporated spatio-temporal context (STC [24]) for head tracking, achieving a prediction of 108 (93.1%) compared to the ground truth of 116 boarding and alighting passengers in bus surveillance videos. Similarly, Zhang et al. [25] employed two variants of the SSD (VGG16-SSD and BMobileNetv3-LiteSSD), in combination with a Kalman filter for head detection and tracking. Their approach was evaluated on CCTV video data from a bus comprising a total of 28 boarding and 81 alighting passengers across seven experiments, as well as on laboratory data including 68 walking-in and 61 walking-out individuals. On the CCTV dataset, the VGG16-SSD-based method achieved a counting accuracy of 100% for boarding and 97.5% for alighting passengers, whereas the BMobileNetv3-LiteSSD-based method obtained 92.7% and 91.4% for boarding and alighting, respectively. On the laboratory dataset, the VGG16-SSD-based approach counted 94.1% walking-in and 88.5% walking-out individuals. Hsu et al. [26] further applied SSD for head and hat detection during the door-opening phase. For tracking, they adopted a three-step cascaded data association scheme- spatial distance, spatial distance and temporal appearance, and motion estimation. Their APC system achieved an accuracy of 94.2%

for boarding and 98.9% for alighting counts across day-time, nighttime, and rainy-day test data, compared with the ground-truth counts of 86 passengers for both boarding and alighting.

Bonis et al. [27] compared two bus passenger counting systems: a combination of SSD and DeepSORT, and a YOLOv5 network integrated into a multi-object tracking (MOT) framework based on Siamese CNNs [28]. The SSD-DeepSORT method estimated 83.9% boarding and 95.0% alighting passengers, while the YOLOv5-Siamese CNN approach counted 94.1% boarding and 92.1% alighting passengers, compared with the ground-truth values of 118 boarding and 239 alighting passengers. More recent studies have applied different versions of YOLO detectors, tiny YOLOv3 [20], YOLOv5 [29], YOLOv5s [30], and YOLOv8 [21], in combination with the DeepSORT tracking method to develop video-based automatic passenger counting (APC) systems, achieving real-time performance. Rawat et al. [21] build two detection models combining Yolo8 trained on the two data set Scut-Head [31] and CrowdHuman [32] with DeepSort. The test was performed on one-hour video data captured by a CCTV camera in a bus including 31 boarding and 12 alighting passengers. On Scut-Head trained APC predicted 22 and 12 and on CrowdHuman trained APC 29 and 13 for boarding and alighting passenger, respectively.

In contrast to abovementioned approaches, DTC pipeline APC systems for 2D video data, we previously introduced an end-to-end neural automatic passenger counting (NAPC) system based on 3D video data [33]. NAPC is based on multi-layer Long Short-Term Memory (LSTM) networks [34] with same number of units each layer. The approach was trained and evaluated on much larger dataset than the studies mentioned before, 31 days of 3D time-of-flight (ToF) video recordings collected on a regional train operating between Berlin and surrounding cities (Berlin-APC dataset [35]) comprising 12,956 door opening videos with 26,243 boarding and 26,164 alighting events. NAPC exploited depth information to identify ticket-relevant passengers, defined as individuals taller than 130 cm. With this design, NAPC directly learned to predict the number of boarding and alighting passengers by minimizing a suitable loss function, thereby enabling an end-to-end learning approach without relying on intermediate detection, tracking, or counting heuristics. The system achieved counting accuracies of 98.5% for boarding and 98.9% for alighting counts.

In related work, Shirazi [36] developed an APC for 3D depth video using a hybrid CNN-LSTM architecture. Trained and evaluated on 62 days of 3D video, the system achieved an overall counting accuracy of 99.6% averaged across boarding and alighting counts. For people counting in retail stores, Nogueira et al. [37] introduced RetailNet, a CNN-based model trained on surveillance video frames using four-channel RGBP inputs, where the fourth channel P is a binary per-pixel mask indicating person visibility. Massa et al. [38] extended this design by appending an LSTM to the CNN (LRCN-RetailNet) to exploit

short-term spatio-temporal coherence. The model processes fixed-length sequences of frames and predicts the count for the final frame. With tuned LSTM hyperparameters, LRCN-RetailNet in the best configuration, obtained 0.365 mean absolute error (MAE) which outperforms both RetailNet (MAE= 1.402) and a YOLOv3 baseline pretrained on COCO [39] (MAE= 4.766).

Our work extends the NAPC framework to enable its application to conventional 2D video data and improve counting accuracy. Because 2D videos lack explicit depth information, either an additional processing stage is required to facilitate height-based passenger counting or the model must discriminate passengers by height-related visual features learned from height-based counting labels during model training. To obtain explicit height-related information, frame-wise monocular depth estimation (MDE) could in principle be used. Recent advances have introduced numerous neural network (NN)-based MDE methods across a wide range of application domains (see reviews in [40], [41], [42], [43], [44]). However, to the best of our knowledge, existing approaches have not been specifically optimized for scenarios such as ours, which involve human height-related analysis in small indoor environments and scenes containing densely distributed, dynamically moving individuals.

To adapt the NAPC method to 2D video data, we therefore adopted the alternative strategy. Instead of integrating an explicit MDE module with an LSTM, we combine convolutional neural networks (CNNs) for spatial feature extraction with LSTM networks for temporal modeling. This CNN-LSTM architecture, referred to as CLAPC, preserves an end-to-end learning and prediction paradigm and is trained on labeled video sequences to robustly identify ticket-relevant passengers defined by a height threshold. Rather than performing explicit height estimation, the network learns a classification-based representation that implicitly captures height-related characteristics encoded in the training labels, enabling reliable passenger discrimination and accurate counting of boarding and alighting events.

To address the undercounting observed in NAPC, we adopt an extended loss function. We trained and evaluated CLAPC on 60 days of 2D video data comprising 25,765 annotated door opening videos with 54,685 boarding and 55,245 alighting events and on the 3D Berlin-APC data. CLAPC achieved counting accuracies of 99.73% (2D) and 99.79% (3D) for boarding and 99.87% (2D) and 99.97% (3D) for alighting counts. Thus, CLAPC applied to 2D video and 3D video data satisfies the $\pm 1\%$ accuracy target of VDV Guideline 457 [45].

The remainder of this paper is organized as follows. Section II presents the 2D video dataset and the preprocessing pipeline. Section III details the NAPC and CLAPC methods and describes the learning procedure, experimental setup, online prediction pipeline of CLAPC, and the performance metrics. Section IV reports CLAPC performance on 2D and 3D datasets and compares NAPC and CLAPC across both datasets. Finally, Section V concludes the paper.

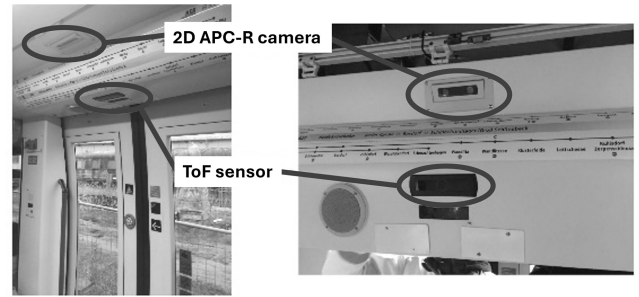


FIGURE 1. Mounting positions of the gray value (2D APC-R) camera and the ToF sensor in the regional train (source: Interautomation Deutschland GmbH).

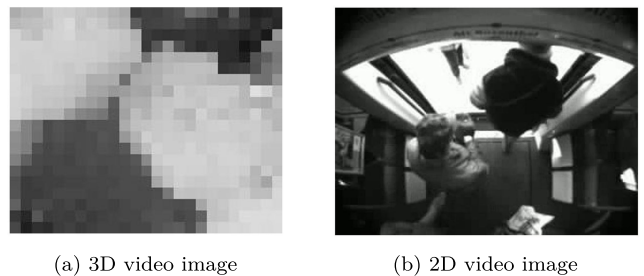


FIGURE 2. Example frames captured by the time-of-flight (ToF) sensor (a) and the 2D APC-R video camera (b) during different door-opening phases. In the 2D video frame (b), two passengers are alighting, while in the 3D video frame (a), two passengers are crossing the door line. For additional details, see Figure 1 in [33]. In (b) only the channel used for the following analyses is shown.

II. DATASETS

We acquired gray value and depth video datasets from an eight-door German regional train operating between Berlin and surrounding cities. A ToF sensor, which generates a 3D depth map, was installed directly above each doorway while a 2D APC-R stereo camera was mounted approximately 30 cm obliquely within the door cove (see Figure 1). Both camera systems were triggered by the door-opening signal and recorded video streams during the door-opening phase. In the following, we refer to each individual video recorded during a single door-opening event as a ‘stream’. The length of each stream varied, as it was governed by the actual door-opening duration and the respective recording frame rate.

We annotated data from 62 days between January and October 2017. The data collected by the ToF sensor was initially recorded at 40 frames per second (fps) with a resolution of 25×20 pixels, but only every fourth frame was retained, yielding an effective frame rate of 10 fps. 31 days of data, referred as to the Berlin-APC dataset, was published, which comprises 12,956 video streams including 26,243 boarding and 26,164 alighting events. Further details on the Berlin-APC dataset can be found in [33] and [35].

The data recorded by the 2D APC-R cameras had a frame rate of 10 fps with a resolution of 320×240 pixels. In our analysis, we used 60 days of gray value data. To mitigate out-of-memory issues during training, we limited the dataset to video streams with fewer than 2,500 frames. This filtering process resulted in a total of 25,767 annotated video streams,

TABLE 1. Statistics of the 2D video dataset. Left: Distribution of the number of frames per stream. Right: Distribution of the number of boarding and alighting events per stream. The dataset comprises 25,767 video streams, including 54,685 boarding events and 55,245 alighting events.

Number of Frames		Number of Events		
range	no. of streams	range	no. of streams	
			boarding	alighting
0–100	94	0	8948	12562
101–200	18303	1	8025	5176
201–300	4677	2–5	6456	5170
301–500	1941	6–10	1369	1636
501–900	619	11–20	685	905
901–1400	93	21–40	262	281
1401–2000	25	41–70	20	37
2001–2500	15			
min.	34	min.	0	0
avg.	204	avg.	2.12	2.14
max.	2445	max.	55	67

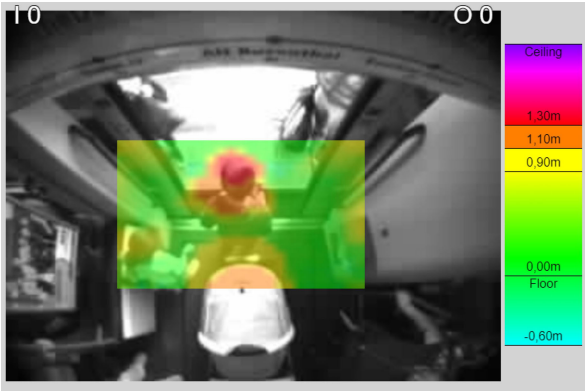


FIGURE 3. Annotators view provided by the annotations software VisualCount. A grayscale 2D video with a resolution of 320×240 pixels is overlaid with a color-coded 3D video with a resolution of 25×20 pixels. See [35] for further details.

encompassing 54,685 boarding events and 55,245 alighting events. An overview of the dataset statistics is provided in Table 1. For the following analyses, we used only one of the camera’s two image streams, thus discarding the stereo information. In the following, we refer to the dataset obtained from the ToF sensor as the 3D (video) dataset or the Berlin-APC dataset, and the dataset acquired from the 2D APC-R stereo camera as the 2D (video) dataset. Figure 2 shows example images captured by the ToF sensor and 2D APC-R video cameras.

The annotations were conducted using the video annotation software VisualCount developed by the company Interautomation Deutschland GmbH.¹ Annotators primarily viewed the 2D video stream, while height information was conveyed through color-coded overlays from the 3D data as individuals crossed the door (see Figure 3). Consequently, the annotation process was performed jointly on both 2D and 3D data. Each video stream was annotated independently by at least three persons, and the number of boarding and alighting

¹<https://www.interautomation.de/en/>

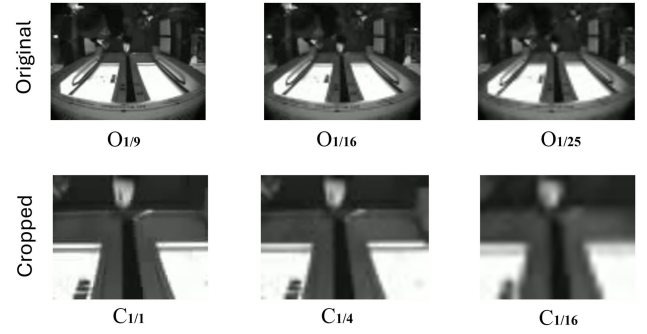


FIGURE 4. Examples of preprocessed 2D video frames. $O_{1/9}$, $O_{1/16}$, and $O_{1/25}$ show the original video frames with spatial resolution decreased by downsampling. $C_{1/1}$, $C_{1/4}$, and $C_{1/16}$ show cropped, door-centered video frames in its original size ($C_{1/1}$) and with decreased resolution ($C_{1/4}$, $C_{1/16}$). The letters ‘O’ and ‘C’ denote original and cropped images, respectively, and the number following each letter indicates the downsampling scale. The corresponding image sizes in pixels are: $O_{1/1}$: 320×240 , $O_{1/9}$: 106×80 , $O_{1/16}$: 80×60 , $O_{1/25}$: 64×48 , $C_{1/1}$: 100×80 , $C_{1/4}$: 54×40 , $C_{1/16}$: 25×20 .

events was determined by majority vote. In cases where the three annotations differed, a fourth and, if necessary, a fifth annotation was conducted to ensure at least two consistent results.

Since we aimed to develop an Automatic Passenger Counting (APC) system capable of determining the number of passengers required to purchase a valid ticket, we followed the fare regulations applied in Germany, where eligibility is primarily defined by age. Because age cannot be directly inferred from video data, height was adopted as a proxy measure. A “passenger” was therefore defined as an individual with a height of at least 130 cm. Individuals with a height of 110 cm or less, corresponding to the average stature of preschool-aged children in Germany, and persons with a height between 110 cm and 130 cm (“critical cases”, reflecting ambiguity in fare classification) were not considered “passengers” in the context of this study. Further details regarding the annotation methodology are provided in [33] and [35].

A. PREPROCESSING OF 2D VIDEO DATA

For the 2D video data, we used one channel of the stereo-video system. To examine the influence of input resolution and spatial context on model performance, we applied six preprocessing strategies involving spatial downsampling and cropping of regions of interest. In three of the preprocessing strategies, the original frames were downsampled to reduce the spatial resolution by factors of 1/3, 1/4, and 1/5 along both the height and width dimensions. In the remaining three strategies, we first extracted the cropped regions, which were centered on the doorway area and corresponded to the region of interest used for the acquisition of the 3D video data. This step allows the model to focus on image regions that are most informative for the passenger counting task. Second, these doorway-centered regions were either used in the original resolution (factor 1) or further downsampled by factors of 1/2 and 1/4 along both spatial dimensions. Visual inspection of the preprocessed data confirmed that spatial downsampling substantially reduced individual identifiability

thus combining reduced computational costs with enhanced privacy protection. Examples of the preprocessed 2D video data are shown in Figure 4.

III. THE CLAPC APPROACH

In our previous work [33], we introduced an end-to-end neural algorithm for APC, referred to as NAPC, which was based on a LSTM architecture. The new approach, termed CLAPC, retains the end-to-end design principle of NAPC but incorporates modifications to the network architecture, most notably the integration of convolutional layers for feature extraction. This section begins with a description of the original NAPC method, which serves as a benchmark, and then presents the CLAPC architecture and training procedure, highlighting the key differences between both approaches. Notably, NAPC and CLAPC rely solely on the annotation results per stream; event times are not used during training.

A. NAPC ALGORITHM

The original NAPC method employs a Long Short-Term Memory (LSTM) neural network to model temporal dependencies. The input layer is fully connected to an intermediate layer, whose number of units matches the number of units in the subsequent LSTM layers. All LSTM layers are of equal size. The final LSTM layer is fully connected to two separate output units, which independently estimate the number of boarding and alighting passengers. Training the NAPC architecture relies on two key features: a bounding-box-based loss function and the concatenation of multiple input streams.

1) BOUNDING-BOX BASED LOSS FUNCTION

If only the numbers of boarding and alighting passengers per stream are available, then we have only two reference points. At the beginning of a stream, no passengers have boarded or alighted, at the end of the stream, the total numbers of boarding and alighting passengers match the annotated target values. Based on this information, we define a bounding box for each passenger type $x \in \{a, b\}$, where a and b denote alighting and boarding. The lower and upper bounds of the bounding box are denoted by $L_{x,i,j}$ and $U_{x,i}$, respectively, where index i refers to the stream and index j , $j \in \{1, \dots, N_i\}$, refers to the frame number in the stream. The total number of frames in stream i is denoted by N_i . The upper bound $U_{x,i}$ of stream i corresponds to the annotated target value for event type x , denoted by $Y_{x,i}$, and remains constant across all frames within stream i . The lower bound $L_{x,i,j}$ of stream i is set to 0 for all frames except the final one, where it equals the current target value $Y_{x,i}$. A predicted value $\hat{Y}_{x,i,j} \in \mathbb{R}$ at frame j in stream i is considered valid and thus excluded from the loss computation, if it lies within the interval $[L_{x,i,j}, U_{x,i}]$. Learning is triggered only when the prediction falls outside this range, indicating either an undercount or an overcount. The bounding-box-based loss for the i -th stream at the j -th

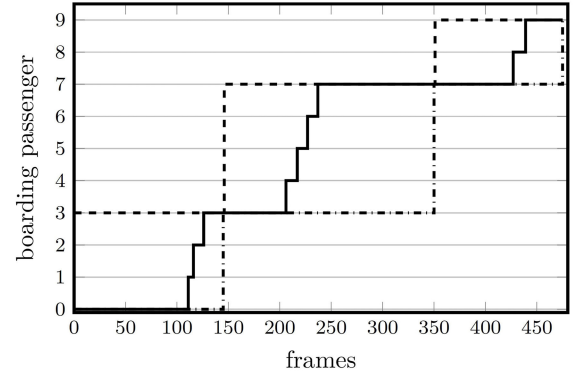


FIGURE 5. Illustration of the upper and lower bounds for counting boarding passenger in an example sequence of three concatenated streams. The dashed (dash-dotted) line represents the upper (lower) bound. The lower bound of the second stream is offset by 3, corresponding to the target count of the first stream. This offset preserves the cumulative state across the sequence, allowing the model to track the accumulated passenger counts over time, as depicted by the jagged black line.

frame and passenger type x is thus defined as:

$$E_{x,i,j} = \underbrace{\max(0, \hat{Y}_{x,i,j} - U_{x,i})}_{\text{overcount}} - \underbrace{\min(0, \hat{Y}_{x,i,j} - L_{x,i,j})}_{\text{undercount}}. \quad (1)$$

2) CONCATENATED SEQUENCES

Using only the loss defined in (1), the LSTM network tends to converge to trivial predictions, most notably, predicting zero across all frames. This behavior arises because the value of zero falls within the allowed bounding box and is never penalized, except for the final frame. To address this limitation and guide the network more effectively toward the correct target values, a strategy based on concatenating multiple video streams was introduced in [33]. Training samples are constructed by combining several individual door-opening phase videos. We refer to each individual video as a ‘stream’ and the concatenated set of streams as a ‘sequence’. For a sequence composed of N_c concatenated streams, we define the cumulative target values:

$$T_{x,i} = \begin{cases} Y_{x,1} & \text{if } i = 1, \\ T_{x,i-1} + Y_{x,i} & \text{if } i > 1, \end{cases}$$

where $x \in \{a, b\}$ and $Y_{x,i}$ denotes the target value of the i -th stream within the sequence. Based on these cumulative targets, we redefine the bounding box for the i -th stream as $[L_{x,i,j}, U_{x,i}]$, where the upper bound remains fixed at $U_{x,i} = T_{x,i}$ for all frames j in the i -th stream. The lower bound is defined as:

$$L_{x,i,j} = \begin{cases} 0 & \text{if } j < F_i, \\ T_{x,i-1} & \text{if } F_{i-1} \leq j < F_i, \\ T_{x,N_c} & \text{if } j = F_{N_c}, \end{cases} \quad F_i = \sum_{t=1}^i N_t,$$

with $i \in \{1, \dots, N_c\}$, where N_i denotes the number of frames in the i -th stream and F_i indicates the cumulative frame count up to the i -th stream within the sequence.

Figure 5 shows the bounding box for a sequence composed of three concatenated streams. Applying this loss function to

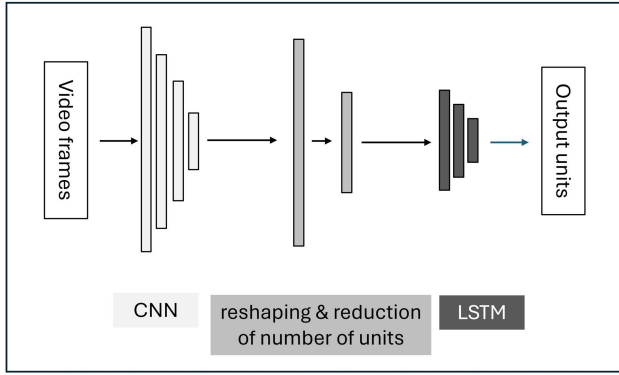


FIGURE 6. The CLAPC network architecture. Each convolutional layer in the CLAPC model is followed by a max-pooling operation to reduce the spatial dimension and retain salient features. The output of the CNN block is subsequently flattened into a feature vector and passed through a fully connected layer with Leaky Rectified Linear Units, where the number of units matches that of the first LSTM layer. This intermediate representation serves as the input to the LSTM network. The LSTM component consists of multiple layers, each with a different number of units, allowing for flexible modeling of temporal dependencies. The final LSTM layer is fully connected to two output units, that predict the target values for boarding and alighting.

concatenated sequences forces the model to learn the intermediate target values to avoid undercounting in subsequent streams. The number of concatenated streams is treated as a hyperparameter during training. In all experiments reported in [33], as well as in the experiments reported here, this parameter was fixed to $N_c = 5$.

3) DATA AUGMENTATION

The training data are augmented at the stream level. Specifically, for each stream in a sequence, it is randomly determined whether to apply horizontal flipping (i.e., mirroring the left and right sides) and/or temporal reversal (i.e., reversing the order of frames). In the case of horizontal flipping, the target values remain unchanged, as the overall passenger counts are unaffected. However, when temporal reversal is applied, the direction of passenger movement is inverted, and the target values for boarding and alighting are swapped accordingly. Temporal reversal also modifies the temporal dynamics and appearance patterns associated with boarding and alighting behaviors, contributing to greater variability in the training data.

B. CLAPC: MODEL ARCHITECTURE

Whereas 3D video data explicitly provides height information, 2D video lacks this cue, which poses a challenge when working with 2D data. To better capture relevant image details, we add a CNN module as a feature extractor prior to the LSTM layers. Figure 6 illustrates the architecture of the CLAPC model. The output of the CNN block is reshaped into a vector and passed through a fully connected representation layer, followed by a Leaky Rectified Linear Unit (LReLU) [46] activation function with a slope parameter α . The number of units in the representation layer is configured to match the dimensionality of the first LSTM layer, ensuring a seamless integration between the convolutional and recurrent components of the model. The output layer consists of two units that predict the target values for boarding and alighting. The number of layers in both the CNN and LSTM blocks,

TABLE 2. Model parameters: Candidate model parameter used for hyperparameter selection. Choices to be evaluated are separated by commas. Numbers in square brackets correspond to one choice and denote properties of the different layers. * (CNN kernels): For an n -layer CNN, use the last n values from the specified lists as the number of kernels per layer. Example: with $[2, 4, 8, 16, 32]$ and $n = 2$, the first and second layers use 16 and 32 kernels, respectively. ** (LSTM units): For an n -layer LSTM, use the first n values from the lists as the hidden units per layer. Example: with $[130, 100, 70, 50]$ and $n = 2$, the first and second layers use 130 and 100 units, respectively. *** (CNN kernel size): Dimensions correspond to the temporal and spatial axes of the input data. The configuration shown preserves temporal resolution while enabling effective spatial feature extraction.

2D data	
CNN no. of layers	2, 3, 4, 5
CNN no. of kernels	$[2, 4, 8, 16, 32]^*$, $[4, 8, 16, 32, 64]^*$
LSTM no. of layers	2, 3, 4
LSTM no. of units	$[100, 70, 50, 30]^{**}$, $[130, 100, 70, 50]^{**}$
batch size	16
buffer size	6
epoch	1500
3D data	
CNN no. of layers	1, 2, 3, 4
CNN no. of kernels	$[8, 16, 32, 64]^*$
LSTM no. of layers	1, 2, 3, 4, 5
LSTM no. of units	50, $[125, 100, 70, 50, 20]^{**}$
batch size	32
buffer size	8
epoch	5000
2D and 3D data	
CNN kernel size	$[1, 2, 2]^{***}$
η	0, 0.2, 0.3, 0.4
slot parameter	0.3
learning rate	0.001

along with the filter and kernel sizes in the convolutional layers, are treated as tunable hyperparameters.

C. EXTENDED BOUNDING BOX

The loss function defined in Section 2 is inherently asymmetric: predictions larger than the target value are penalized more severely than those smaller than the target value. This asymmetry arises because overcounting is defined as predicting values greater than the target, whereas undercounting is evaluated relative to a lower bound—typically smaller than the target value, except in the final frames of each stream. Consequently, minimizing this loss function encourages predictions to remain between the lower bound and the target value, while penalizing any prediction that exceeds the target throughout the stream (except in the final frame). This behavior can lead to a model that slightly but systematically undercounts. [33]. To mitigate this effect, we introduce an extended bounding box interval $[L_{x,i,j}, U_{x,i}]$, whose upper bound $U_{x,i}$ is increased beyond the target, $T_{x,i}$, i.e., $T_{x,i} + \eta$, by some value $0 < \eta < 0.5$. This adjustment promotes a more balanced penalty for over- and underestimations during training, thereby reducing the model's tendency to systematically undercount.

D. LEARNING PROCEDURE AND ENVIRONMENT

To ensure an unbiased evaluation across training, we employed 10 distinct training-validation-test splits. Each split consists of all streams, respectively, from a randomly selected

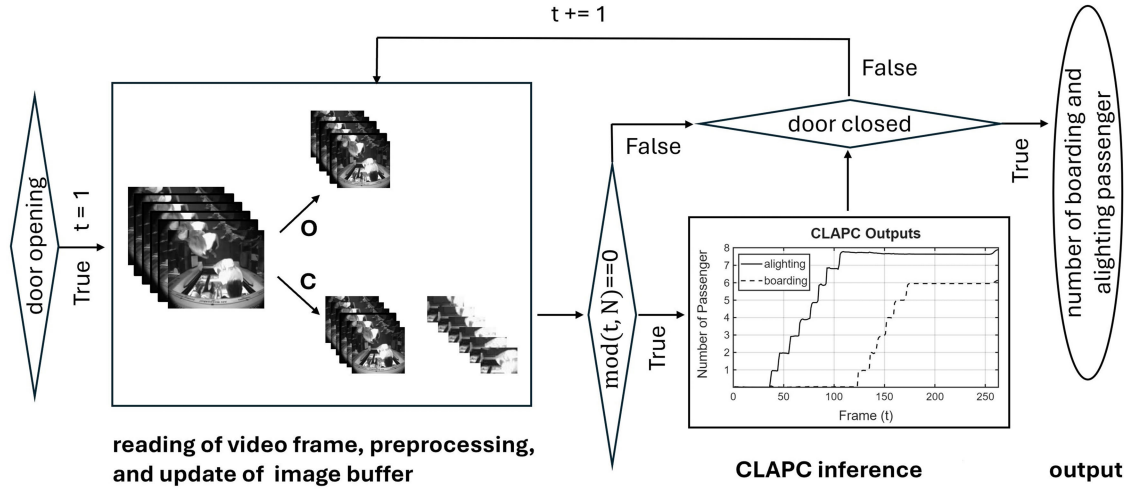


FIGURE 7. Online prediction pipeline of CLAPC. While the vehicle door is open, incoming video frames are processed sequentially. RGB frames are converted to grayscale, whereas grayscale frames are used directly as single-channel inputs. Each frame is then down-sampled (O) or down-sampled and cropped (C), followed by normalization through division by 255. The processed frame is appended to an image buffer, which serves as input to the CLAPC model. Inference is performed on every N -th frame ($N \geq 1$, $N \in \mathbb{N}$). For $N > 1$, computational cost is reduced, enabling real-time operation and accelerating prediction. The cycle of frame capture, preprocessing, buffer update, and CLAPC inference continues as long as the door remains open. After the door closes, the final CLAPC outputs are rounded to the nearest integer and reported as the estimated numbers of boarding and alighting passengers.

set of 36 training days, 12 validation days, and 12 test days. The model is trained on the 36-day training set, and the best-performing model is selected based on its performance on the corresponding validation set. Final performance metrics are then computed on the respective test set. All results reported in this section are derived from the outcomes of these 10 test scenarios. We train the CLAPC model using the Adam optimizer [47] for both the 2D and 3D datasets. Table 2 summarizes the candidate parameters used in our experiments, distinguishing between those specific to 2D and 3D data and those shared between them.

All experiments were implemented using AlmaLinux 8.10, CUDA Driver Version 12.6, Runtime Version 12.0, TensorFlow 2.6, Python 3.9.15, and NumPy 1.19.5. Computations were performed on the high-performance computing cluster of the TU Berlin, Faculty of Mathematics, which is optimized for batch-processing workloads.²

E. ONLINE PREDICTION PIPELINE

The trained CLAPC model can be deployed for online prediction on the same train type from which the training data were collected. Figure 7 illustrates the flowchart of the online prediction pipeline. Upon detection of a door-opening event, video frames are sequentially captured, preprocessed, and appended to a data buffer serving as input to the CLAPC model. Inference is performed on every N -th frame ($N \geq 1$, $N \in \mathbb{N}$); choosing $N > 1$ reduces the computational load and improves the processing speed. Preprocessing and inference continue iteratively until a door-closing event occurs. The final CLAPC outputs are subsequently rounded to the nearest integer and reported as the predicted numbers of boarding and alighting passengers.

F. PERFORMANCE METRICS

Counting performance is evaluated using three different metrics which assess different aspects of the counting task and which are described below in the subsections entitled ‘Classification Metric’, ‘Regression Metric’, and ‘Counting Metric’. In the following, the true number of passengers and the number of passengers predicted on the test set are denoted by $Y_{x,i}^k$ and $\hat{Y}_{x,i}^k$, where $x \in \{b, a\}$ denotes boarding (b) and alighting (a) passengers. k is the index of the data split, and i is the index of stream for the classification and regression metrics. For the counting metric, i denotes the index of a set of door opening phases associated with a specific door at a stop within a train schedule. In this case, the corresponding target value and prediction for the i -th set are defined as the sum of the true and predicted values, respectively, over all individual streams within that set. Real-valued model outputs are rounded to the nearest integer to obtain $\hat{Y}_{x,i}^k$.

1) CLASSIFICATION METRIC

Each video stream is assigned to a class labelled by the number of boarding or alighting passengers. A prediction is correct when the number matches the number for that stream. The classification accuracy in the k -th test set is, therefore, defined as:

$$ACC_x^k = 100 \times \frac{1}{N_k} \sum_{i=1}^{N_k} \delta(Y_{x,i}^k = \hat{Y}_{x,i}^k) \quad (2)$$

where N_k is the number of streams in the k -th data split in the 10-fold training-validation-test scenario and $\delta(\cdot)$ is the indicator function, which returns 1 if its argument is true and 0 otherwise. ACC_x^k thus represents the percentage of video streams in the k -th test dataset for which the number of boarding or alighting passengers is estimated correctly. Classification accuracy, however, ignores the magnitude of

²<https://www.tu.berlin/en/hpc-cluster>
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the prediction error. As a result, predicting two boarding passengers for a video stream with three actual boardings is treated the same as predicting two passengers for a stream with 100 actual boardings.

2) REGRESSION METRIC

To account for the magnitude of the counting error, we also evaluate the performance using two commonly adopted regression error metrics: Mean Absolute Error (MAE) and Symmetric Mean Absolute Percentage Error (SMAPE). These metrics are defined as follows:

$$\text{MAE}_x^k = \frac{1}{N_k} \sum_{i=1}^{N_k} |Y_{x,i}^k - \hat{Y}_{x,i}^k| \quad (3)$$

$$\text{SMAPE}_x^k = \frac{1}{N_k} \sum_{i \in C} \frac{|Y_{x,i}^k - \hat{Y}_{x,i}^k|}{0.5 \times (|Y_{x,i}^k| + |\hat{Y}_{x,i}^k|)} \times 100,$$

$$C = \{i : (|Y_{x,i}^k| + |\hat{Y}_{x,i}^k|) \neq 0\} \quad (4)$$

where MAE_x^k and SMAPE_x^k denote the test performance of the model on the k -th data split in the 10-fold training-validation-test scenario. MAE provides an average of the absolute differences between the predicted and the true values of the target variable, offering an intuitive measure of prediction accuracy. SMAPE, on the other hand, expresses those differences as a percentage and symmetrically normalizes deviations relative to both predicted and actual values. This dual-metric evaluation provides both an absolute and a relative perspective on the prediction error, allowing for a more nuanced assessment of the model's generalization capabilities.

3) COUNTING METRIC

In addition to these classical machine learning metrics, we evaluate the system according to two domain-specific criteria defined in the VDV Guideline 457 [45], which was developed by the Association of German Transport Companies. These criteria include the global relative bias (GRB) and the confidence interval derived from an equivalence test based on a two one-sided t-test (TOST equivalence test). Both metrics are essential for evaluating the compliance of APC systems with established industry standards.

The GRB measures the overall counting error. It quantifies the deviation between the predicted and the true number of boarding or alighting passengers summed over all streams in the test set. It is defined as

$$\begin{aligned} \text{GRB}_x^k &= 100 \times \frac{\sum_{i=1}^{N_k} (\hat{Y}_{x,i}^k - Y_{x,i}^k)}{\sum_{i=1}^{N_k} Y_{x,i}^k} \\ &= 100 \times \left(\frac{\sum_{i=1}^{N_k} \hat{Y}_{x,i}^k}{\sum_{i=1}^{N_k} Y_{x,i}^k} - 1 \right) \end{aligned} \quad (5)$$

A negative GRB (i.e., $\sum_{i=1}^{N_k} \hat{Y}_{x,i}^k < \sum_{i=1}^{N_k} Y_{x,i}^k$) indicates undercounting, whereas a positive GRB (i.e., $\sum_{i=1}^{N_k} \hat{Y}_{x,i}^k > \sum_{i=1}^{N_k} Y_{x,i}^k$) indicates overcounting. As a descriptive measure, GRB reflects the average counting bias of the model.

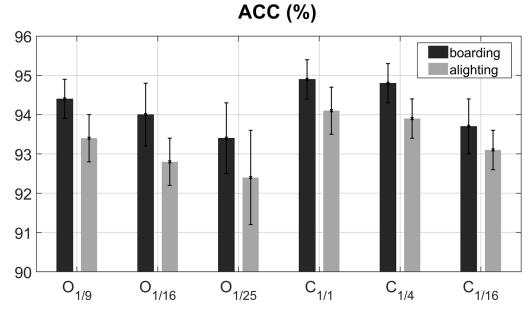


FIGURE 8. Classification performance of the CLAPC method on the six 2D video datasets described in Fig. 4. The figure shows the mean and the standard deviation of the percentage of correctly classified streams across ten test sets. Black (gray) bars represent boarding (alighting) events.

Because the signed GRB averaged over 10 test sets can mask sizeable errors—over- and underestimations may balance out—we also consider the absolute GRB (aGRB), which is defined as

$$\text{aGRB}_x^k = 100 \times \frac{\left| \sum_{i=1}^{N_k} \hat{Y}_{x,i}^k - Y_{x,i}^k \right|}{\sum_{i=1}^{N_k} Y_{x,i}^k}. \quad (6)$$

The equivalence test based on the TOST procedure assesses whether the confidence interval (CI) of the aGRB lies entirely within a predefined equivalence margin $[-\Delta, +\Delta]$. If the CI lies within these bounds, the aGRB is statistically insignificant concerning the equivalence margin, thereby supporting the hypothesis of practical equivalence between predicted and manual passenger counts. We computed the 95% confidence interval (CI) based on a TOST-induced equivalence testing procedure as described in [45].

IV. RESULTS

We experimented with various model configurations (Table 2) and evaluated their performance using the classification, regression, and counting metrics (see Section III-F). In Section A, we present the results for the proposed CLAPC approach on the six differently preprocessed 2D video datasets. In Section B, we compare the performance of the proposed CLAPC model with that of NAPC and CLAPC using the original bounding boxes on the data set C_{1/4}. The corresponding model parameters are provided in Table 3. This comparison aims to evaluate: (i) the effectiveness of a hybrid CNN-LSTM model for height-based object detection compared to an LSTM-only approach, and (ii) the impact of extended bounding boxes in reducing the undercounting tendency. Section C presents a comparison between CLAPC and NAPC on the 3D dataset, highlighting the differences between the two approaches when explicit height information is available.

A. PERFORMANCE OF CLAPC ON 2D VIDEO DATA

Fig. 8 shows the mean and standard deviation of the classification accuracy for counts of both boarding and alighting passengers. The mean classification accuracy lies in the range

TABLE 3. Optimal parameters of the CLAPC architecture determined by hyperparameter selection for the six datasets described in Fig. 4 and optimal model architecture of NAPC and CLAPC trained using the original bounding box for data set $C_{1/4}$.

	3D data	$O_{1/9}$	$O_{1/16}$	$O_{1/25}$	$C_{1/1}$	$C_{1/4}$	$C_{1/4}^{ob}$	$C_{1/4}^{napc}$	$C_{1/16}$
CNN no. of layers	3	4	3	3	4	4	3		3
CNN no. of kernels	[16,32,64]	[4,8,16,32]	[8,16,32]	[8,16,32]	[4,8,16,32]	[8,16,32,64]	[8,16,32]		[8,16,32]
LSTM no. of layers	3	3	3	4	3	3	3	4	3
LSTM no. of units	[125,100,70]	[130,100,70]	[130,100,70]	[100,70,50,30]	[130,100,70]	[100,70,50]	[100,70,50]	[50]*	[100,70,50]

The superscripts 'ob' and 'napc' denote the CLAPC approach with the original bounding box and the NAPC method, respectively. The n -th values in the square brackets indicate the number of units (for LSTM) or kernels (for CNN) in the n -th layer. For example, [125, 100, 70] denotes an LSTM with 3 layers of 125, 100, and 70 units; [16, 32, 64] denotes a CNN with 3 layers of 16, 32, and 64 kernels. [50]* means [50,50,50,50].

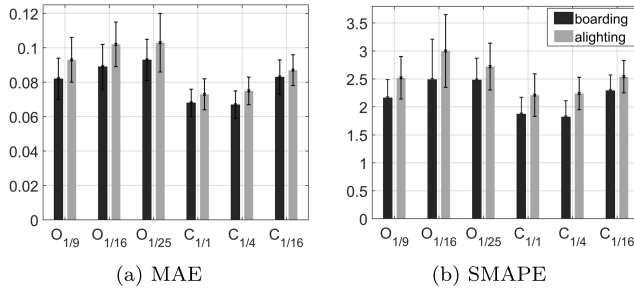


FIGURE 9. Average counting errors of the CLAPC method on the six 2D video datasets described in Fig. 4. Mean and standard deviation of the Mean Absolute Error (MAE) (a) and the Symmetric Mean Absolute Percentage Error (SMAPE) (b) across the 10 test sets. Black (gray) bars represent boarding (alighting) events.

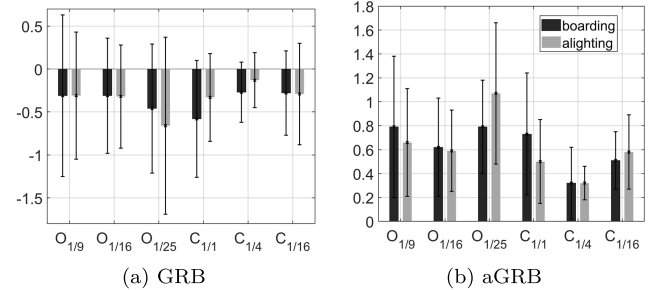


FIGURE 10. Accuracy of passenger counts summed over all streams in a test set for the six 2D video datasets described in Fig. 4. The figure shows the mean and the standard deviation of the GRB (a) and the aGRB (b) across the 10 test sets. Black (gray) bars represent boarding (alighting) events.

of [93.4%, 94.9%] for boarding and [92.4%, 94.1%] for alighting with mean boarding accuracies consistently exceeding alighting accuracies. The average accuracy computed over the cropped datasets is slightly higher (94.5% for boarding and 93.7% for alighting) than over the datasets containing the entire image (93.9% for boarding and 92.9% for alighting). Furthermore, less resolution leads to decreased accuracy for both the whole-image and cropped-image datasets. Of all the datasets, $C_{1/1}$ and $C_{1/4}$ exhibit the best classification performance.

Fig. 9 shows the mean and standard deviation of the Mean Absolute Error (MAE) and the Symmetric Mean Absolute Percentage Error (SMAPE) when predicting the number of boarding and alighting passengers. The MAE ranges between 0.07 and 0.11, while the SMAPE lies between 1.83% and 3.0%. Consistent with the classification results, regression performance is slightly better for boarding compared to alighting events for both metrics and across all six datasets. Furthermore, applying CLAPC to the cropped images results in slightly improved accuracies for both regression metrics compared to using full-frame images. Of all the datasets, $C_{1/1}$ and $C_{1/4}$ exhibit the best performance for both MAE and SMAPE.

Fig. 10 shows the accuracy of passenger counts summed over all streams in the test set. The mean values of the Global Relative Bias lie between -0.66% and -0.13% , corresponding to a counting accuracy between 99.34% and 99.87%. This shows that the CLAPC system consistently achieves high precision in passenger counting across various

preprocessing settings. The mean values, however, still consistently lie below zero, indicating a systematic tendency toward slight undercounting. Unlike the general trends observed for the other metrics, prediction of boarding only outperforms prediction of alighting counts for dataset $O_{1/25}$. For datasets $O_{1/9}$, $O_{1/16}$, and $C_{1/16}$, boarding and alighting performances are comparable. Conversely, datasets $C_{1/1}$ and $C_{1/4}$ show superior performance for alighting passenger counts compared to boarding counts.

The mean aGRB values are larger in magnitude than their signed GRB counterparts but remain below 1% in every scenario except for the alighting count task on dataset $O_{1/25}$. Notably, the CLAPC system achieves both its smallest GRB (-0.27% for boarding and -0.13% for alighting) and its lowest aGRB (0.32% for both, boarding and alighting) for dataset ($C_{1/4}$), reflecting the most accurate and least biased performance across all datasets.

Fig. 11 shows the average 95 % confidence intervals (CI) of the aGRB computed based on the TOST equivalence testing procedure (see Section 3). For all datasets, the lower bounds of the CIs for both boarding and alighting are greater than -1% , but an upper bound below 1% is only achieved for dataset $C_{1/4}$. According to the VDV Guideline 457 [45], an Automatic Passenger Counting system must satisfy specific accuracy requirements. In particular, the system must achieve a GRB in the range $[-1\%, 1\%]$. Furthermore, the 95% confidence interval of the aGRB, as determined using a Two One-Sided Tests (TOST) equivalence testing procedure, must

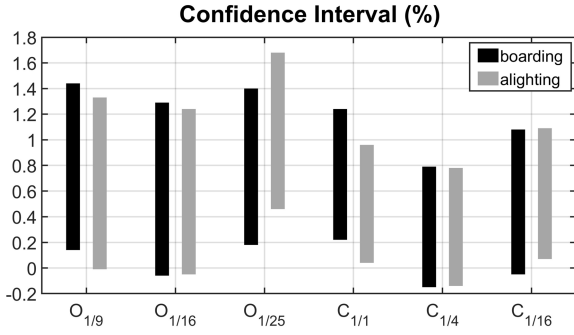


FIGURE 11. Average 95% confidence intervals (CIs of the aGRB) computed using a TOST-based equivalence testing procedure. Each interval is defined by the mean of the lower and upper CI bounds across the 10 test sets. Results are shown separately for the six 2D video datasets described in Fig. 4. Black (gray) bars represent boarding (alighting) events.

be entirely contained within the equivalence margin of $[-1\%, 1\%]$. As shown in Fig. 11, dataset ($C_{1/4}$) meets these criteria on average, with its confidence interval being fully contained within the specified equivalence range. Above results also show, that the optimal counting performance is achieved, when the image is cropped to show the area under the door, which contains the information most relevant for boarding and alighting. Hence choosing the right image region can further improve the accuracy of APC systems.

B. COMPARISON CLAPC AND NAPC ON 2D DATA AND IMPACT OF THE EXTENDED BOUNDING BOX ON MODEL PERFORMANCE

In this section, we evaluate the improvement in height-based passenger counting achieved by the CNN module of CLAPC and the effect of training with the extended bounding box cost function. Experiments were conducted on the dataset $C_{1/4}$. NAPC was tested with three to six LSTM layers, each containing 50 units, while CLAPC was evaluated using both original and extended bounding boxes under the same experimental settings. The optimal model parameters are summarized in Table 3. Fig. 12 shows the mean and standard deviation of the five performance metrics and the 95% confidence interval from an equivalence test, across all 10 test runs, comparing NAPC, CLAPC with the original (CLAPC-ob), and the extended bounding box (CLAPC-eb).

CLAPC consistently outperformed NAPC across all six performance metrics. Paired t-tests at a significance level of $\alpha = 0.005$ further confirmed that CLAPC-ob achieved significant improvements over NAPC for both boarding and alighting counts. Averaged over 10 test runs, the mean performance (averaged over boarding and alighting passenger counts) improved by approximately 5% in classification accuracy (from 88.5% to 93.5%), by over 50% for MAE (from 0.176 to 0.087) and SMAPE (from 5.66% to 2.74%), and by about 78% for GRB (from 5.67% to 2.73%). On the 2D dataset (without depth information), height-based passenger classification demands more advanced object recognition than a standalone LSTM can achieve. The observed performance

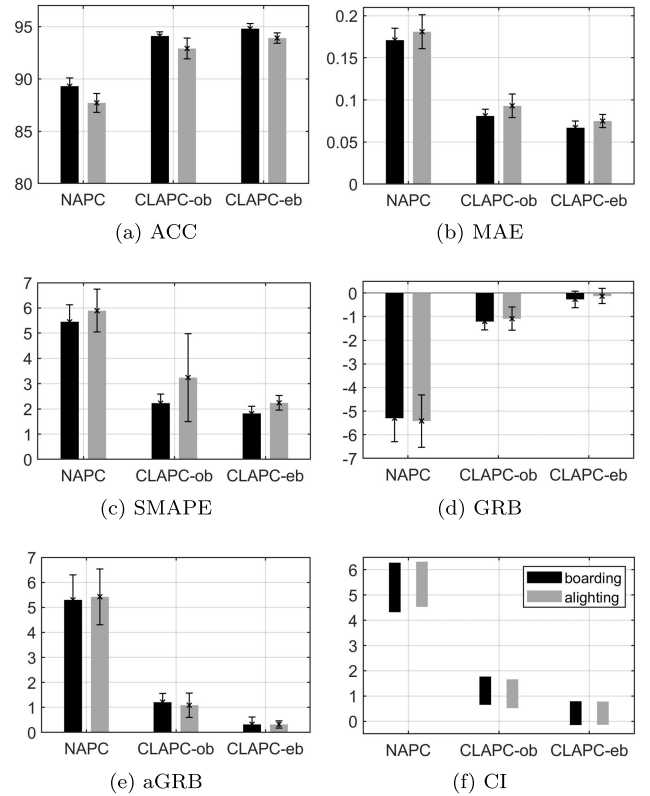


FIGURE 12. Performance comparison between NAPC, CLAPC using the original (CLAPC-ob), and CLAPC with extended bounding box (CLAPC-eb), trained on the dataset $C_{1/4}$. Model parameters are summarized in Table III.

gains demonstrate that the CNN-LSTM hybrid effectively meets this requirement.

Furthermore, CLAPC with extended bounding boxes (CLAPC-eb) significantly outperformed its original bounding box variant (CLAPC-ob) based on paired t-tests at a significance level of $\alpha = 0.005$, except for ACC in both boarding and alighting and SMAPE in alighting. The most substantial gains were observed in counting accuracy. Undercounting was reduced by 77% for boarding (from -1.21 to -0.27) and by 88% for alighting (from -1.09 to -0.13). These improvements enable CLAPC-eb to meet the counting accuracy requirements of the VDV 457 Guideline, demonstrating the effectiveness of the extended bounding box in mitigating systematic undercounting.

C. COMPARISON CLAPC AND NAPC ON 3D VIDEO DATA

In this section, we evaluate the performance of NAPC and CLAPC on a dataset containing height information. CLAPC is compared with the NAPC using a five-layer LSTM architecture with 50 units per layer using the Berlin-APC dataset. The NAPC results are taken from [33]. Table 4 summarizes the average performance of both methods, showing that CLAPC consistently outperforms NAPC across all evaluation metrics. The improvements include a 1.1% (1.6%) increase in classification accuracy for boarding and alighting, respectively, along with substantial reductions in MAE (38% for boarding and 31% for alighting). Furthermore, consistent

TABLE 4. Average classification, regression, and counting performance of NAPC and CLAPC on the Berlin-APC dataset.

Performance of NAPC and CLAPC on 3D data				
Metric	Boarding		Alighting	
	NAPC	CLAPC	NAPC	CLAPC
ACC (%)	96.2	97.0	95.4	97.0
MAE	0.060	0.042	0.055	0.037
GRB (%)	-1.46	-0.21	-1.11	-0.03
CI ^(lower) (%)	-2.24	0.02	-1.70	-0.15
CI ^(upper) (%)	-0.68	1.24	-0.53	0.74

The NAPC performance values are taken from Table 2 in [33], based on eight training-validation-test splits, while the CLAPC results are based on ten such splits.

with the results on 2D data, CLAPC markedly reduces the undercounting bias observed in NAPC on 3D data, as indicated by the improvement in GRB from -2.24% to -0.06% for boarding and from -1.11% to -0.17% for alighting. However, these improvements for 3D data remain considerably smaller than those obtained when both methods are applied to 2D data (see Section B, Fig. 12). Importantly, the NAPC system failed to meet the VDV 457 requirements for both GRB and CI. In contrast, CLAPC satisfied the GRB criterion on average and fulfilled the CI requirement for alighting. For boarding, the upper bound of the confidence interval was 1.08%, only marginally exceeding the 1% threshold. Notably, on 3D data, CLAPC-eb and CLAPC-ob achieved nearly identical performance across all evaluated metrics; therefore, we report only the results of CLAPC-eb. This suggests that, for 3D data, the hybrid CNN-LSTM model alone is sufficient to achieve performance gains, whereas the more tolerant loss function plays a crucial role in improving passenger detection on 2D data.

V. SUMMARY AND DISCUSSION

In this work, we introduced CLAPC, a hybrid CNN-LSTM architecture for real-time automatic passenger counting using conventional 2D video data acquired from a top-down perspective. Building upon the NAPC framework, we integrated CNNs for spatial feature extraction, LSTMs for temporal modeling, and an extended bounding-box-based loss formulation. CLAPC estimates boarding and alighting passenger counts, where passengers are defined as individuals exceeding a height threshold of 130 cm.

To optimize performance on 2D video data, six preprocessing strategies were evaluated, including a reduction of resolution and door-centered cropping. These steps reduce computational cost while enhancing privacy protection. Among the tested configurations, the approach $C_{1/4}$ achieved the best results in terms of global relative bias (GRB) and confidence intervals. Accordingly, grayscale input combined with door-centered cropping is recommended for 2D APC deployment. CLAPC achieved mean GRB values of 99.73% for boarding and 99.87% for alighting. Strong performance was also observed on 3D video data, with accuracies of

99.79% (boarding) and 99.97% (alighting). CLAPC satisfies the VDV 457 requirements on 2D data with respect to both GRB and confidence intervals. On 3D data, all GRB criteria and the confidence interval requirement for alighting were met, while the boarding confidence interval marginally exceeded the threshold (1.08%).

Comparative evaluation across all six metrics — classification accuracy, MAE, SMAPE, GRB, absolute GRB, and confidence intervals — demonstrates that CLAPC consistently outperforms NAPC on both 2D and 3D datasets. The improvements are particularly pronounced for 2D data, underscoring the effectiveness of CNN-based spatial feature extraction in the absence of explicit depth information.

CLAPC is an end-to-end neural-network-based Automatic Passenger Counting (APC) system that can implicitly extract passenger height information from 2D video data for counting purposes and that does not require separate learning modules, e.g. for depth estimation, object detection, tracking, or counting. By adopting an end-to-end learning paradigm, the system avoids interdependencies between multiple processing stages, thereby simplifying the overall system design. The proposed approach can in principle directly leverage existing surveillance camera infrastructure for passenger counting. Compared to depth-video-based APC systems, 2D CCTV-based APC eliminates the need for additional hardware installation and provides a cost-effective alternative to 3D video-based solutions that rely on dedicated depth sensors.

Adapting the system to new vehicles may require the collection of additional training data. The end-to-end learning framework, however, requires only the numbers of boarding and alighting passengers per video stream as labels. 3D information is necessary for annotations based on passenger height, but this information can be obtained, for example, by using an additional Time-of-Flight sensor during the data collection period which can later be removed. During inference, CLAPC estimates the numbers of boarding and alighting passengers using only 2D video data and with better performance than the original 3D-based approach. This serves as a proof of principle that 3D information can be reliably estimated from 2D video data in the counting context.

The proposed CLAPC method is applicable not only to passenger counting, including individuals who are not required to purchase tickets, but also to the counting of other object categories, such as bicycles and baby buggies, using 2D video data. The current stream-based labeling approach requires the concatenation of multiple video streams, which leads to high GPU memory consumption due to the resulting long training sequences. By adopting an event-based labeling strategy, learning can be performed on individual events rather than concatenated streams, eliminating the need to combine multiple sequences into a single training sample. This has the potential to significantly reduce GPU memory usage. Furthermore, extending the method to robustly count additional object categories represents an interesting direction for future research.

ACKNOWLEDGMENT

The authors would like to thank the research department staff of Interautomation Deutschland GmbH, Berlin, for their domain expertise and for providing the 3D and 2D video recordings. Open access publication fees were covered by German Science Foundation and the Open Access Publication Fund of the Technische Universität Berlin.

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